

Abhiram Yanamandra

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“Dynamic Software Engineer with 2+ years’ experience in Full-Stack Development, Cloud-Native Applications, and CI/CD automation”

EDUCATION

University of New South Wales, Sydney, NSW

Mar. 2022 – Aug. 2026

Bachelor of Engineering in Computer Engineering (Honours)

WAM: 68.6

- Redback Racing Member – 2nd in 2023 and 4th in 2025
- Student Ambassador, Engineering Dean’s Unit
- Software Engineering Fundamentals – 89 HD
- Frontend Engineering – 86 HD
- Object Orientated Design and Programming – 74 CR

Sep. 2023 – Jun. 2025

Apr. 2025 – Present

SUMMARY

Computer Engineering student with 2+ years of experience in embedded systems, full-stack development, and cloud-native applications. Proven track record in building automated data pipelines and visualisation solutions that drive business acumen and operational excellence. Adept at transforming complex datasets into actionable insights using SQL, Python and cloud platforms like Microsoft Azure and AWS to support strategic decision-making.

SKILLS

- **Programming & Scripting:** C, C++, Java8+, Python, TypeScript, JavaScript, SQL, Bash/Shell, HTML5, CSS3, JSON, Kotlin, C#
- **Frameworks & Libraries:** React, NodeJS, NextJS, ShadcnUI, TailwindCSS, Microservices, Pandas, OpenCV, NumPy
- **Database Technologies:** MySQL, PostgreSQL, NoSQL (Elasticsearch, MongoDB), Redis
- **Cloud & DevOps:** AWS (Lambda, DynamoDB, EC2, S3), GCP, Microsoft Azure, Docker, Kubernetes, Terraform, Jenkins
- **Testing & Tools:** JUnit, Jest, Playwright, Cypress, Postman
- **Data Analytics & Visualisation Tools:** Power BI, Kafka, Prometheus, Grafana,
- **Soft Skills:** Motivated, Communication Skills, Strong Problem-Solving Skills, Willingness to Learn, Committed, Friendly, Ambitious
- **Methodologies & Deployment:** Agile/Scrum, TDD, Pair Programming, Code Reviews, Jira, Confluence, Bitbucket, GitHub

PROFESSIONAL EXPERIENCE

Test Engineering Intern, Lightspeed Photonics, Singapore

Jan. 2025 – Feb.2026

- Researched methods and assessed feasibility for automating visual inspection of optical components including VSCELs, Photodiodes, Optical Lenses, Fibers and Assemblies, enhancing **technical competence in QA automation and ensuring quality standards**.
- Developed and evaluated **machine learning and image-processing pipelines using Python, OpenCV, Pandas and NumPy** for automated visual inspection and dimensional accuracy checks.
- Documented inspection methodologies, workflows, and evaluation results for repeatability and knowledge transfer.
- Executed product testing workflows using advanced test benches and **data analysis tools** to ensure rigorous quality assurance.

Data Analyst & Full-Stack Developer, UNSW Redback Racing, Sydney

Sep. 2023 - Jun. 2025

- Experience in all the technical phases of **Software Development Lifecycle** including Requirement Analysis, Design, Implementation and Testing. Built and enhanced features for a large-scale systems including a Live Telemetry Website using **ReactJS, NextJS, NodeJS, TypeScript and Tailwind CSS**, shipping production ready which improved overall car performance.
- Introduced **Automated Testing** using **Playwright** which increased test coverage by 20%, ensuring **reliability and security of critical data reporting** systems.
- Utilised **Docker and AWS for cloud computing solutions**, ensuring **robust and scalable** performance for telemetry systems.
- Implemented a centralized logging system for vehicle sensor data, which converted each car setup run into structured logs and storing them in **S3 buckets for long term analysis and traceability**. This enabled cross-run and cross-configuration comparisons by preserving historical logs, which allowed race engineers to perform **data analytics** and optimise car setups based on telemetry data.

Training Clerk, Singapore Armed Forces, Singapore

Jan. 2020 - Jan. 2022

- Designed and implemented a **Power BI dashboard to visualize** training competencies and physical fitness metrics of **more than 100+ soldiers**, enabling commanders to **identify performance gaps** and allocate training resources more effectively, earning **commendation** from the unit **Commanding Officer**.
- **Streamlined** fitness test administration and emergency response coordination by **standardizing** reporting procedures, reducing administrative processing time and **improving data accuracy across training cycles**.

PROJECTS

AI Pest Detector: UNSW COMP9517

- Trained a **YOLOv8m** model to detect and classify pests from the AgroPest-12 dataset.
- Model achieved **precision of 91.7%, 85.1% recall and 91.9% mAP@50**, which highlights model’s capability to accurately detect and classify pests.

Smartwatch Prototype: UNSW DESN2000

- Designed firmware for a wearable smartwatch prototype displaying simulated metrics in real-time on a 16x2 LCD.
- Implemented custom CGRAM glyphs for heart rate and battery metrics, focusing on **effective data visualisation** for end users.
- Prototyped the screen UI on figma to aid in development process.